

Hawler Medical University



Ascaris lumbricoides, Trichuris trichura, Strongyloides stercoralis, Trichinella spiralis

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Member: In European Pharmaceutical Review.

In Ifad platform KSA.

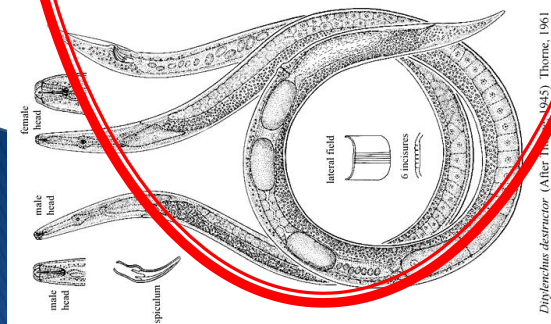
Reviewer: Walailak Journal of Science and Technology.

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HELMINTHOLOGY

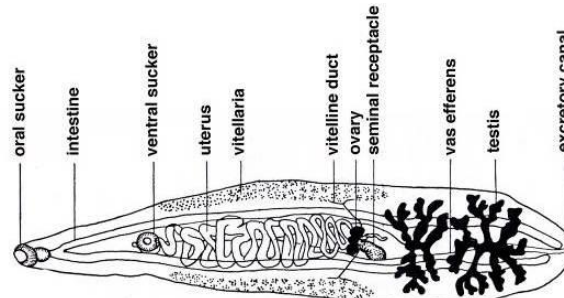
Nematodes (Round Worm)

Unsegmented
Posses mouth, oesophagus,
anus
Gen. Sexes separate



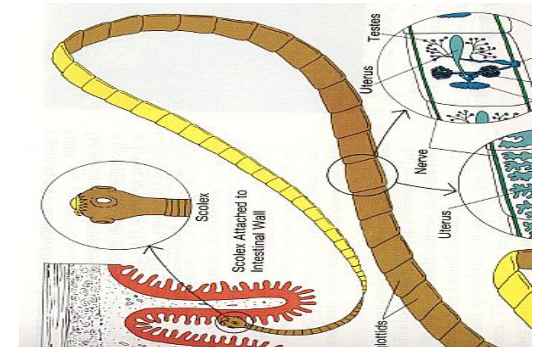
Trematodes (Flukes)

Unsegmented,
Leaf-like or cylindrical
Gen. hermaphroditic and
eggs with lids (exp Schisto)
Min two hosts, one is snail
Have suckers
In liver, intestine, lung &
blood



Cestodes (Flatworm)

Segmented,
Posses scolex, neck,
proglottids
Gen. hermaphroditic
Have suckers / hooks that
grasp the host
Gen. eggs have hooks, no
lids



NEMATODES

Intestinal

- ▶ *Ascaris lumbricoides*
- ▶ *Trichuris trichura*
- ▶ *Enterobius vermicularis*
- ▶ Hook worms
 - *Ancylostoma & Necator*

Tissue

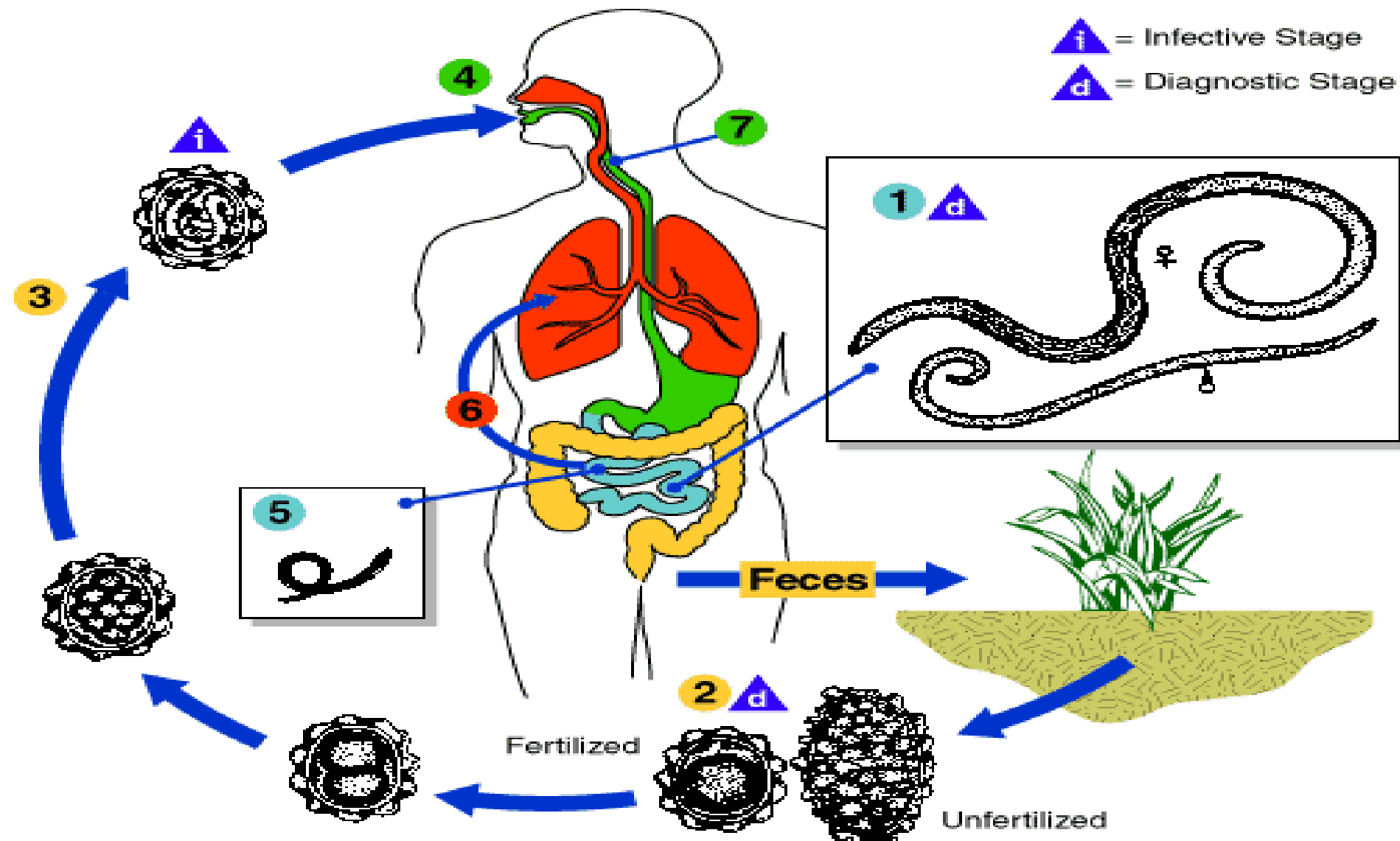
- ▶ *Strongyloides stercoralis*
- ▶ *Trichinella spiralis*
- ▶ Larva migrans
 - *Toxocara sp.*
- ▶ *Dracunculus medinensis*

Blood

- ▶ Filariasis
 - *Wuchereria bancrofti*
 - *Brugia malayi*
 - *Onchocerca volvulus*
 - Others

Ascaris lumbricoides (Ascariasis)

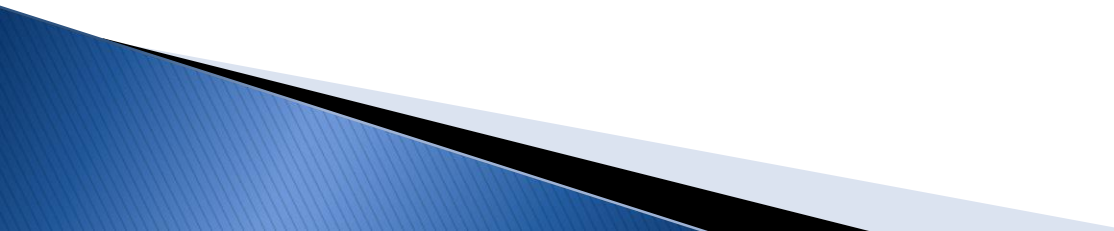
i = Infective Stage
d = Diagnostic Stage



Transmission & Epidemiology

- ▶ Unfertilized eggs are not infective
 - ▶ Fertile eggs embryonate and become infective after several weeks, depending on the environmental conditions (moist, warm).
 - ▶ After infective eggs are swallowed, the larvae hatch, invade the intestinal mucosa, and are carried via the portal, then systemic circulation to the lungs
 - ▶ The larvae mature further in the lungs, penetrate the alveolar walls, ascend the bronchial tree to the throat, and are swallowed
- 

Transmission & Epidemiology

- ▶ Upon reaching the small intestine, they develop into adult worms
 - ▶ Adult worms live in the lumen of the **small intestine** up to 2 years
 - ▶ They reach a size of 40 cm
 - ▶ A female may produce ~ 200,000 eggs/day
 - ▶ Definitive host is human, no intermediate host
- 

Transmission & Epidemiology

- ▶ The most common infection in humans
- ▶ Worldwide distribution, one billion/year infected
- ▶ Highest prevalence in tropical and subtropical regions, and areas with inadequate sanitation
- ▶ More prevalent in children (mostly 5-9 years old)
- ▶ **Soil**-transmitted infection
- ▶ human swine are the major hosts for *Ascaris*
- ▶ Transmission depends on the dissemination of **eggs** in environmental conditions suitable for their **maturation** (by finger, water, foods contaminated by soil contact)

Clinical Features

Intestinal phase of adult worms

- ▶ Usually cause no acute symptoms
- ▶ High worm burdens may cause abdominal pain and intestinal obstruction, constipation



Massive Ascaris infection in child. A large bolus of roundworms expelled following anthelmintic treatment.



Clinical Features

Intestinal phase of adult worms

- ▶ May migrate other parts of the body cavities and it blocking of the biliary tract, appendicitis, or nasopharyngeal expulsion.

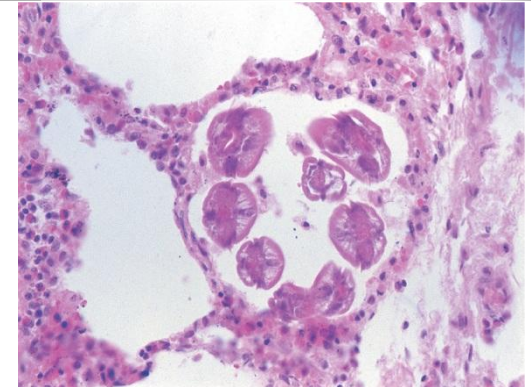


Ascaris lumbricoides in appendix

Clinical Features

Lung phase of larval migration

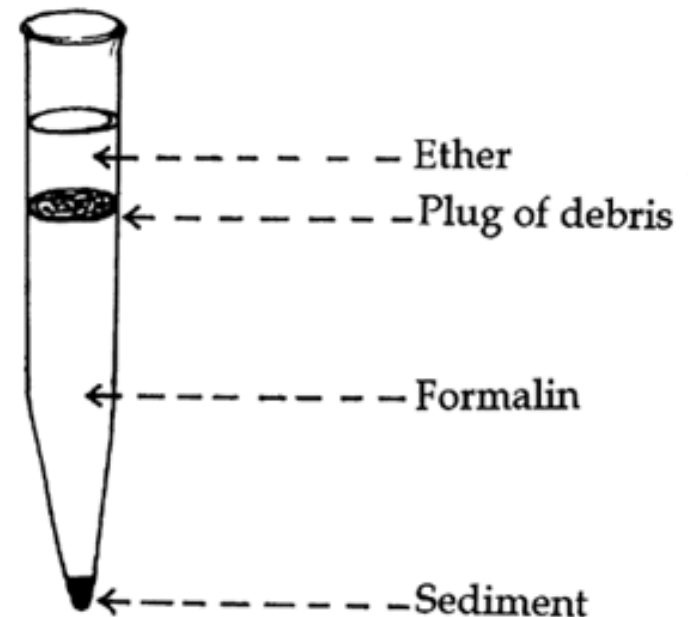
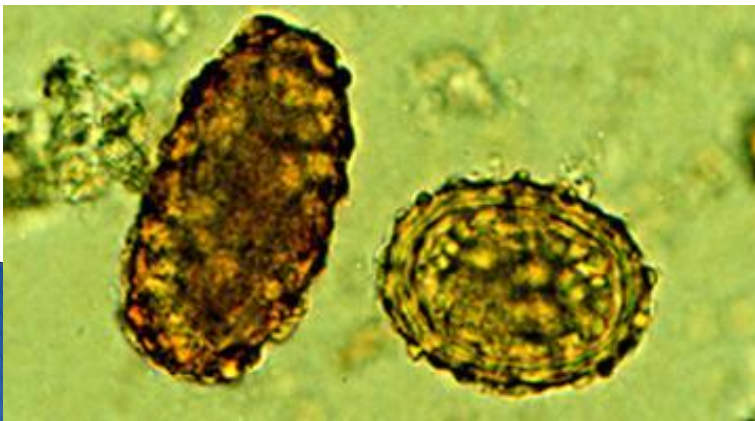
- ▶ Pulmonary symptoms can occur (cough, dyspnea, hemoptysis, eosinophilic pneumonitis)
Löeffler Syndrome
- ▶ Wheeze and bronchospasm due to allergic reactions
- ▶ Eosinophilia

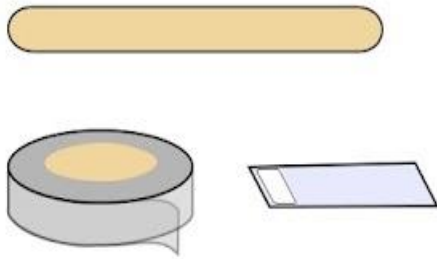


Transverse sections of *Ascaris* larvae in pulmonary alveoli

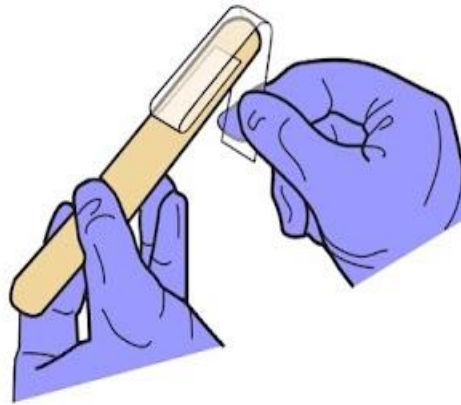
Diagnosis

- ▶ Diagnosis based on microscopy
- ▶ Finding eggs (fertilized/unfertilized) or adults in faeces
- ▶ Using concentration method is better
- ▶ Rarely larvae can be found in sputum
- ▶ Scooch tap

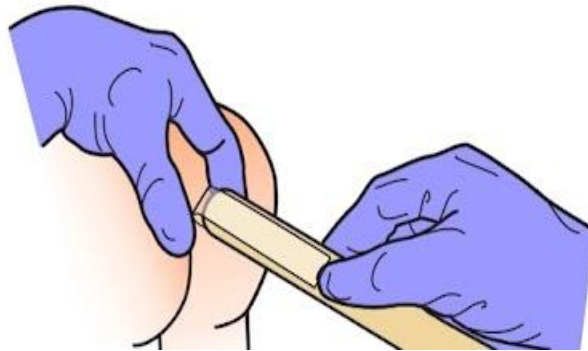




1. Pita perekat (transparan),
stick es krim, dan objek glass



2. Dengan selotip dilingkarkan (sisi berperekat ke arah luar)
di atas penekan lidah dari kayu

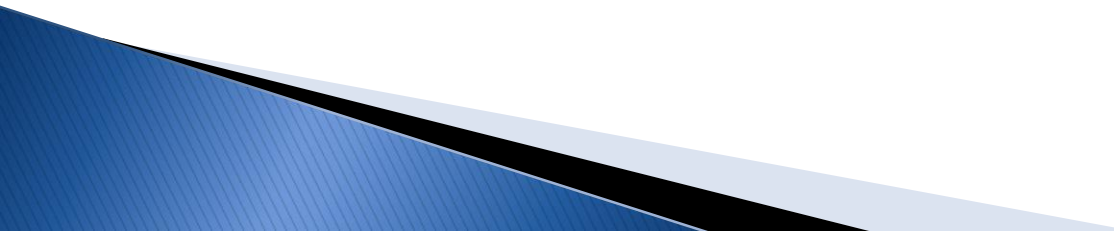


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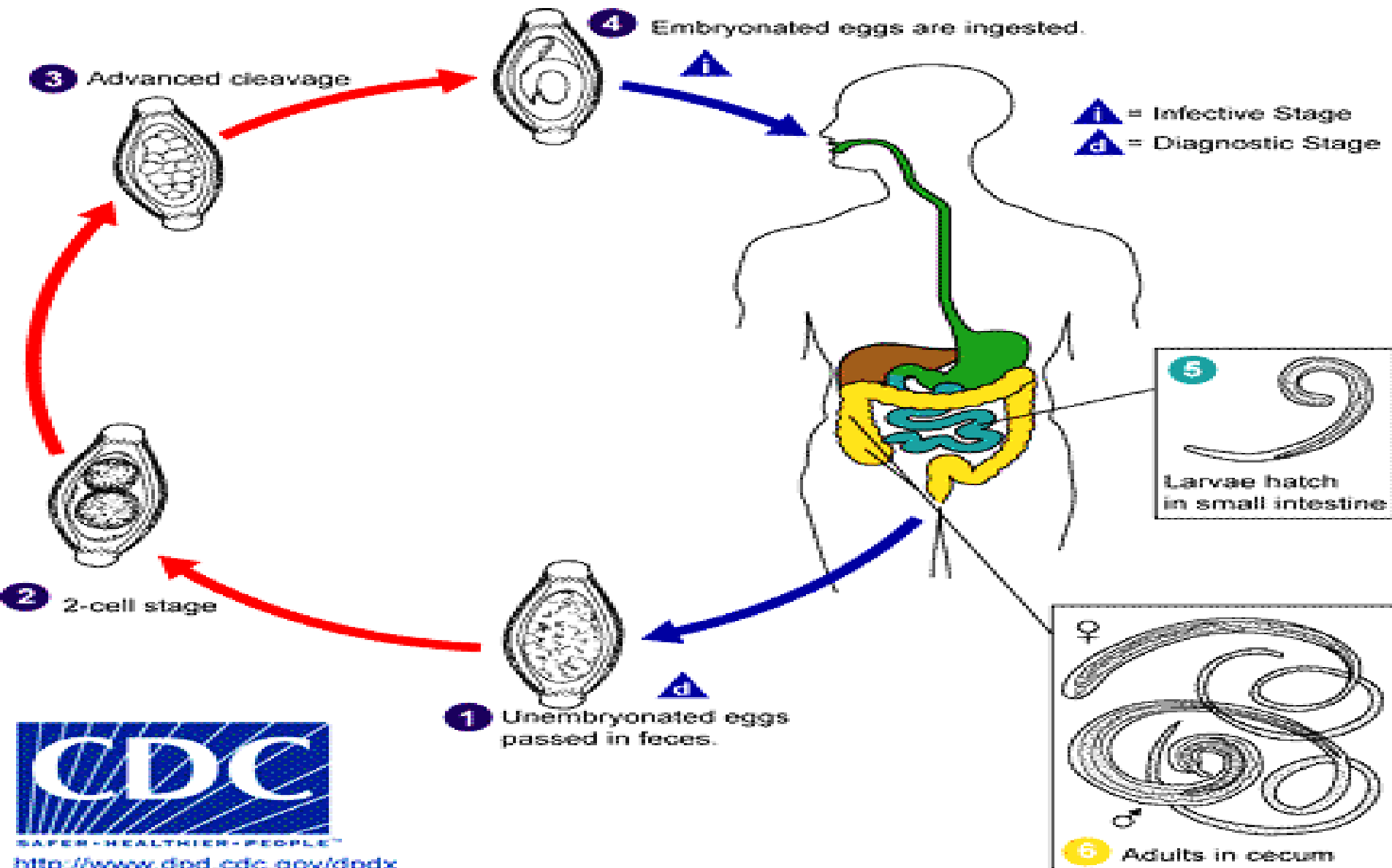
Pengambilan harus dilakukan
pada pagi hari sebelum pasien
mandi atau ke kamar mandi.



Treatment & Prevention

- ▶ Pyrantel pamoate (can be used in pregnancy),
Mebendazole or Albendazole
 - ▶ Sanitary practice in fecal disposal
 - ▶ Good hygienic practices with food
 - ▶ Regular hand washing
 - ▶ Mass de-worming programmes
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Trichuris trichura (Trichuriasis)

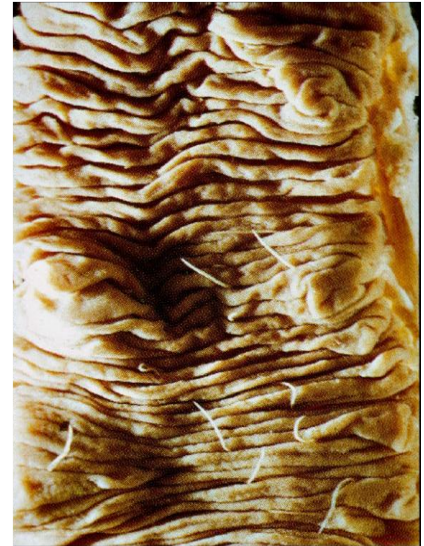


Transmission & Epidemiology

- ▶ The non-embryonated eggs are passed with the stool
- ▶ In the **soil**, eggs embryonate and become infective in 15-30 days
- ▶ After ingestion (soil-contaminated hands or food), the **eggs** hatch in the small intestine, and release larvae that **mature** and establish themselves as adults in the colon
- ▶ Usually they are together with *Ascaris* (because of soil transmission)

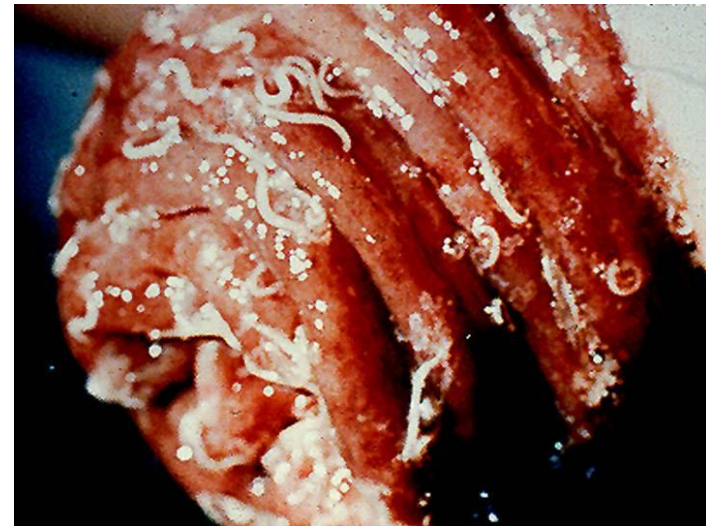
Transmission & Epidemiology

- ▶ The adult worms (~ 4 cm) live in the **colon**
- ▶ Also known as «whipworm»
- ▶ Fixed with the anterior portions threaded into the mucosa (the head is in the end of the "tail")
- ▶ Female worms shed 3,000-20,000 eggs/day



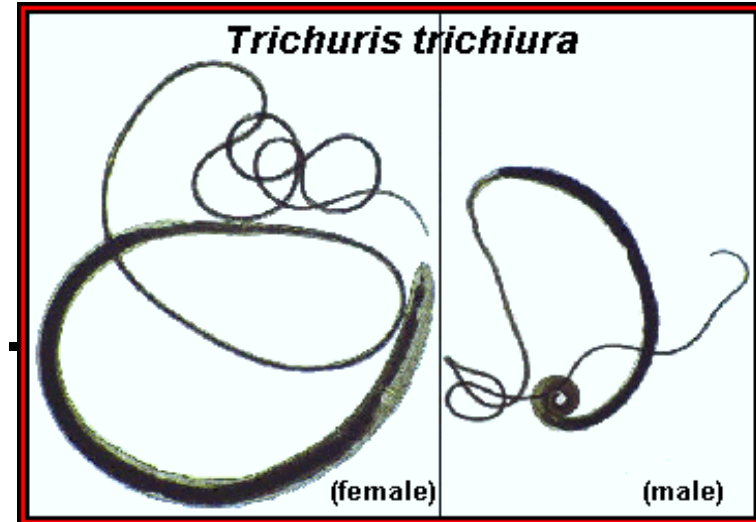
Clinical Features

- ▶ Most frequently asymptomatic
- ▶ Heavy infestations cause symptoms especially in small children
- ▶ GIS problems (tenderness, abdominal pain, chronic mucoid & bloody diarrhea)
- ▶ Weight loss, weakness, growth retardation
- ▶ Anemia and iron deficiency from blood loss possibly growth retardation.
- ▶ Rectal prolapse



Diagnosis & Treatment & Prevention

- ▶ All same as *Ascaris*
- ▶ Diagnosis based on microscopy
- ▶ Finding eggs (lemon shaped) or adults (whipworms) in faeces
- ▶ Using concentration method is better
- ▶ Examination of the rectal mucosa by proctoscopy (or directly in case of prolapses)
- ▶ Treatment is highly effective and can last 1–3 days.
- ▶ Dosage guidelines are the same for adults and children.

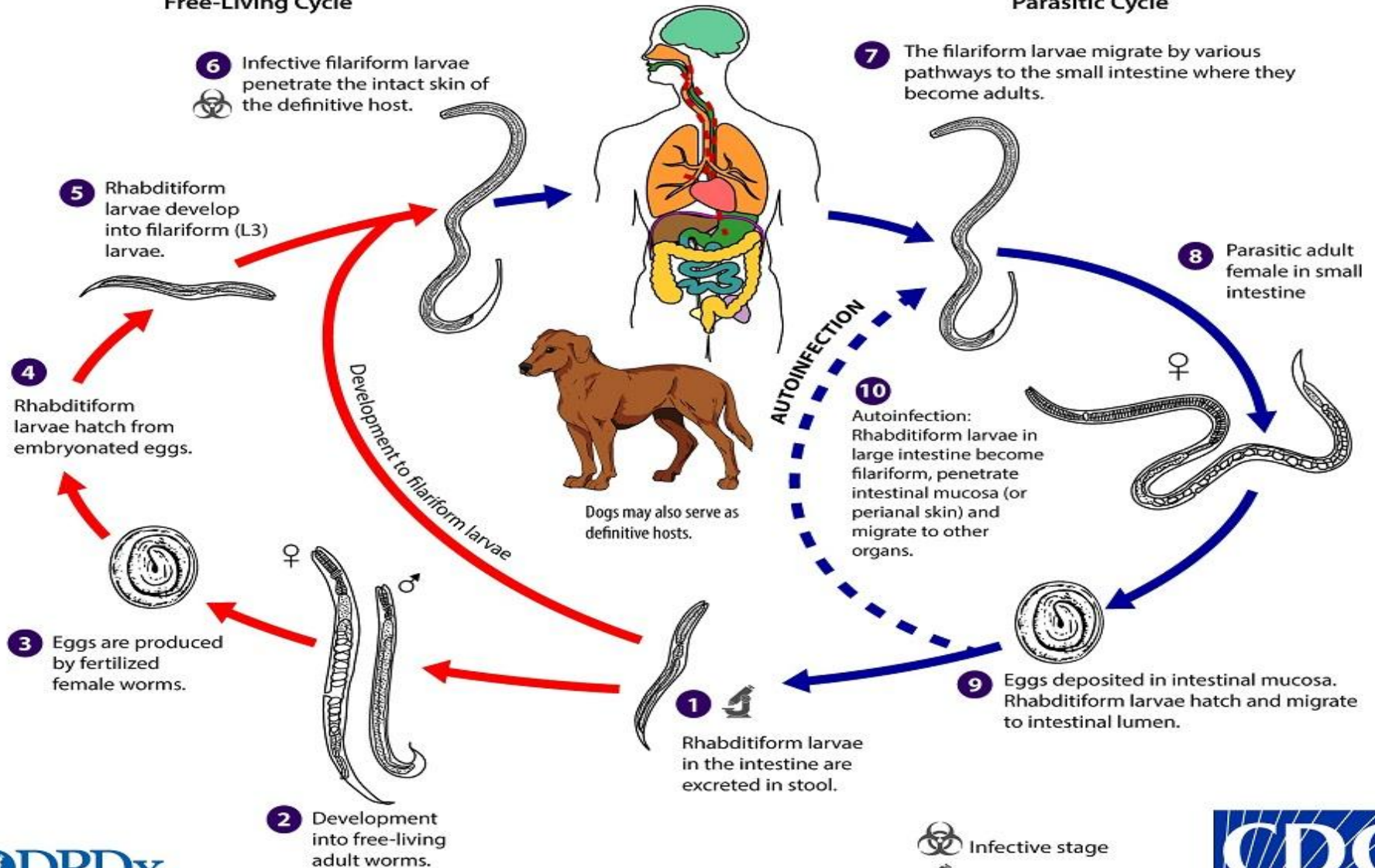


(Strongyloidiasis)

Strongyloides stercoralis

Free-Living Cycle

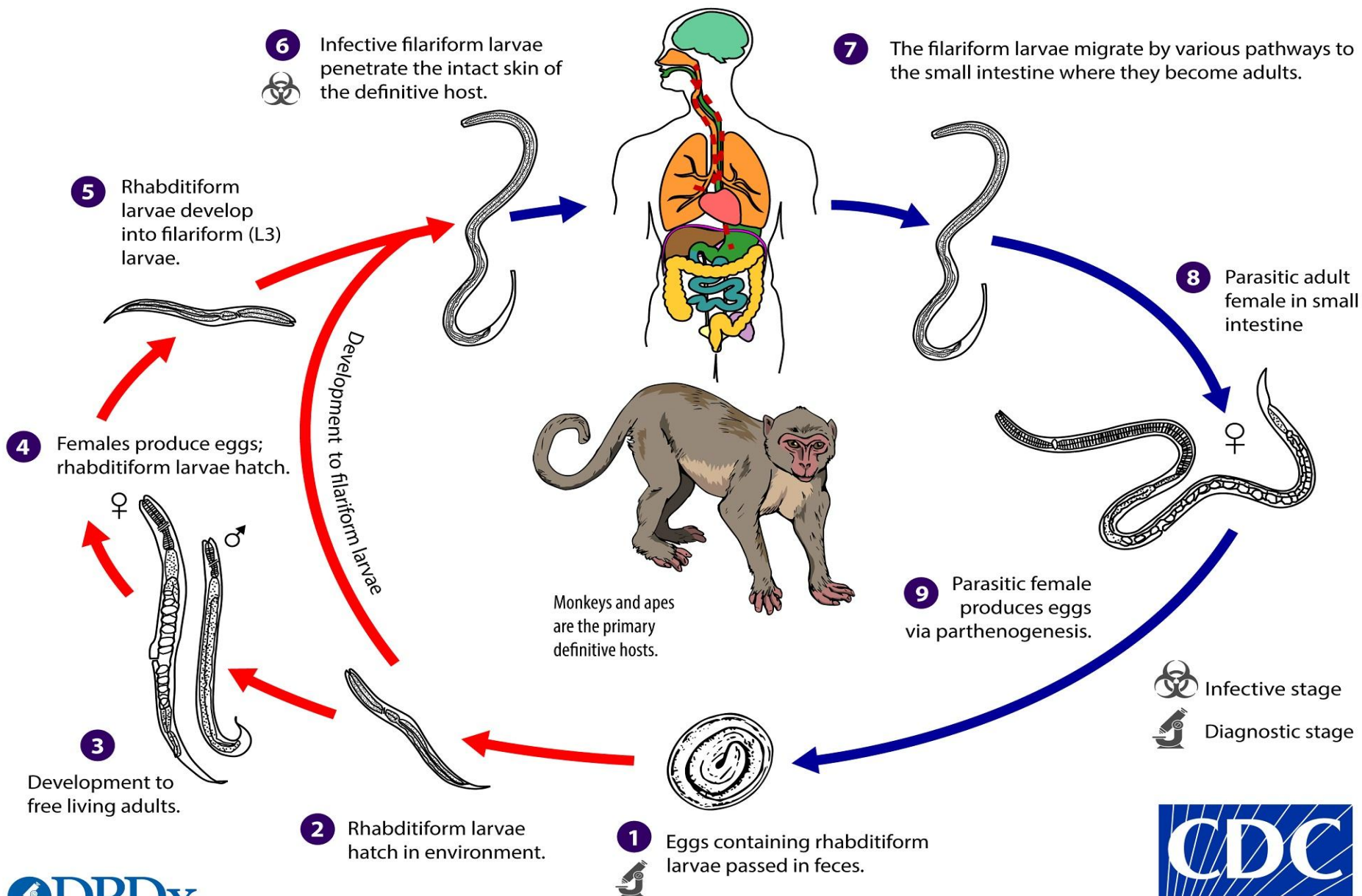
Parasitic Cycle



Strongyloides fuelleborni

Free-Living Cycle

Parasitic Cycle

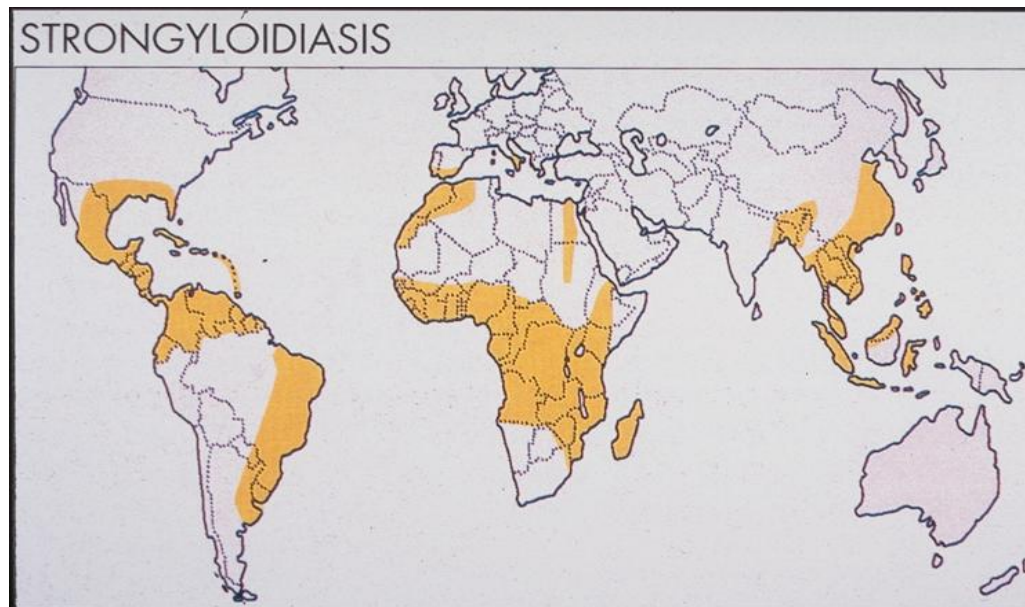


Transmission & Epidemiology

- ▶ Very complicated life cycle, there are some free living forms in the soil
- ▶ Rhabditiform larvae are passed out in faeces
- ▶ They become **filariform larvae** which is infective
- ▶ This transformation take place either sexually with male and female in the soil or asexually in the intestine (**autoinfection**)
- ▶ They reach to small intestine via blood stream
- ▶ Only adult females live embedded in the small intestinal mucosa and reproduces in a "**Parthenogenetic** manner"

Transmission & Epidemiology

- ▶ Called as threadworm
- ▶ Soil-transmitted diseases
- ▶ Worldwide prevalence: ~100 million



Clinical Features

- ▶ Skin penetration may cause local irritations: rash, pruritus
- ▶ Larva currens is seen on the trunk and buttocks
- ▶ Migrating larvae can cause mild respiratory symptoms in a few week with asymptomatic eosinophilia (Löffler pneumonitis)
- ▶ Sometimes ectopic larvae can be found in the brain or other viscera



Clinical Features

- ▶ Established infection may have no sign
- ▶ In severe infection manifestations include intestinal symptoms: epigastric pain, abdominal distension, vomiting, diarrhea with voluminous stools and malabsorption with dehydration and electrolyte disturbance, anemia
- ▶ Hyper-infection results from massive autoinfection in the presence of immunosuppression
 - hemorrhagic diarrhea, intestinal perforation
 - hemorrhagic pneumonia, gram-negative meningitis, shock, sepsis with high mortality.
 - Up to 75% of people with chronic strongyloidiasis have mild peripheral eosinophilia or elevated IgE

Diagnosis & Treatment

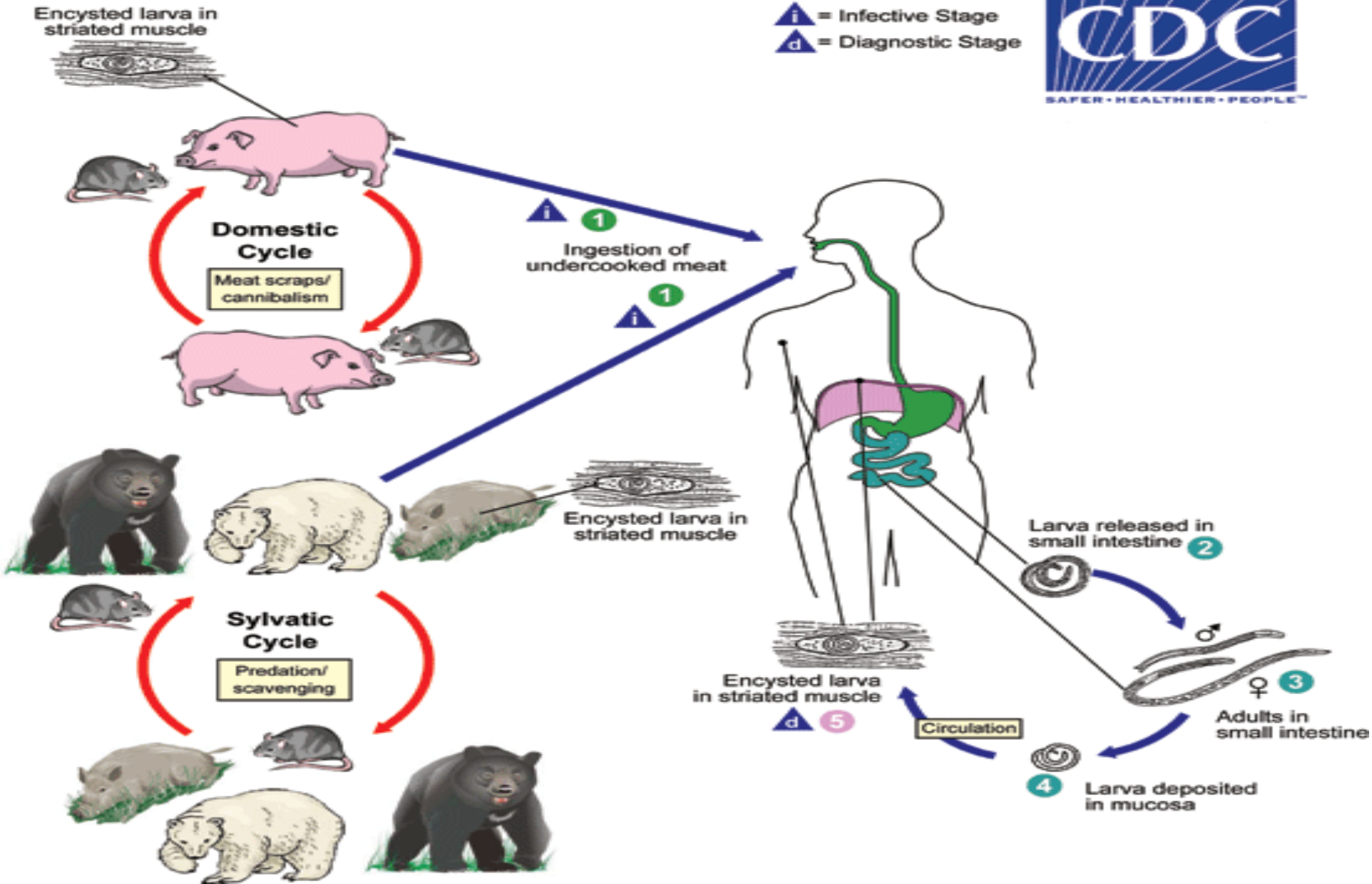
- ▶ Rhabditiform larvae in faeces or duodenale aspiration or sputum
- ▶ Mebendazole or thiabendazole



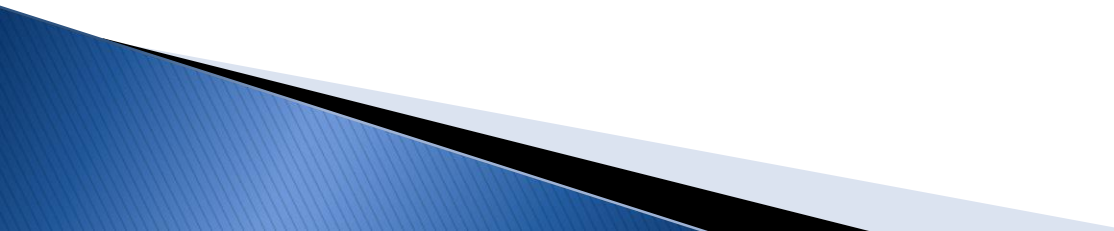
Trichinella spiralis (Trichinellosis)

Encysted larva in striated muscle

i = Infective Stage
d = Diagnostic Stage



Transmission & Epidemiology

- ▶ Zoonotic infection, worldwide
 - ▶ Usually human infection is contracted from the domestic pigs
 - ▶ Larvae penetrate the intestinal wall and are carried into the muscle cells where they coil up
 - ▶ Become infective within 3 weeks
 - ▶ If the larvae are ingested by a new host in undercooked meat, they are released and develop into adults around a day.
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Transmission & Epidemiology

- ▶ Adults live in the intestine, larvae in the muscle
- ▶ Viviparaous
- ▶ Infected pork ingestion is main source
- ▶ Food-borne epidemics can be seen



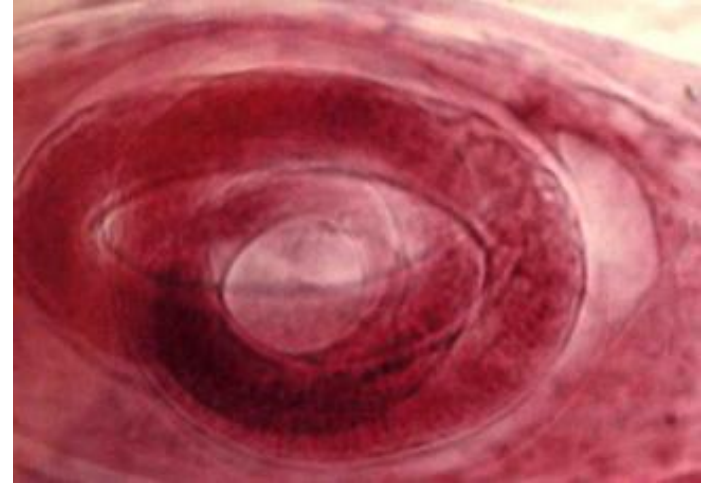
Clinical Features

- ▶ Intestinal inflammation leading to diarrhoea.
- ▶ Migration of larvae provide inflammatory response leading to periorbital oedema, haemorrhages under nails, muscle pains and myocarditis (1week)
- ▶ Especially muscles of respiration and tongue are involved
- ▶ Eosinophilia is high
- ▶ In long term eventual fibrosis and degeneration, resulting in calcification
- ▶ conjunctivitis, myalgias



Diagnosis & Treatment

- ▶ Clinical features
- ▶ Microscopic examination
- ▶ Serology
- ▶ Muscle biopsy



Treatments:

- ▶ Mebendazole or Thiabendazole
- ▶ Steroid

